

Habnetic Paper v2

Revision plan after RSS - September 2026

Working basis: *Posterior-Based Decision Stability under Cross-City Transfer in Urban Flood Risk Prioritisation*. The objective is to sharpen the decision problem, remove methodological ambiguity, and make the experiments harder to attack without turning this manuscript into Phase 4.

1. Rewrite the abstract and opening around the decision problem

Make the paper answer the question raised at RSS immediately: **why does this matter?** The central distinction should be that uncertainty in estimated risk is not the same as uncertainty in the resulting decision. For a municipality with capacity to inspect or intervene on only k assets, the relevant question is whether uncertainty changes which assets cross that selection boundary. Frame the output operationally as robustly-in assets, robustly-out assets, and a small review set requiring additional evidence or judgement.

2. Replace Figure 1 with the model that is actually fitted

The current conceptual diagram implies a latent hazard process with hazard parameters, while the implemented experiment uses simplified hazard and exposure proxies in a Bayesian logistic model. Redraw the figure as the actual computational pipeline: raw spatial/environmental data $\rightarrow$ exposure and hazard proxies $\rightarrow$ Bayesian logistic model $\rightarrow$ posterior risk draws $\rightarrow$ ranking per draw $\rightarrow$ top- k membership $\rightarrow$ decision-stability metrics. Keep the latent-hazard architecture for future work rather than implying that it is already part of the present model.

3. Specify the synthetic outcome generation completely

Document the full data-generating process for the synthetic Bernoulli outcome: coefficients, intercept, feature transformations, random seed, event-rate logic, and whether the same generating mechanism is used across cities. Explain why those settings were selected. The manuscript currently states that the synthetic outcome isolates decision behaviour from calibration, but reproducibility requires the exact generating mechanism.

4. Clarify what 'cross-city transfer' actually means

State precisely what is transferred from Rotterdam to Hamburg and Donostia-San Sebastian. If the same specification is fitted independently in each city, use language such as **cross-city stress test under fixed specification**. If posterior parameters are actually transferred, describe that mechanism explicitly. Also quantify covariate distribution shift rather than inferring stronger shift only from a wider decision boundary.

5. Report instability relative to the decision capacity, not only the whole city

Keep borderline share as a proportion of all assets, but add more operational denominators. At the current approximately 1% thresholds, the reported shares correspond to roughly 30 borderline buildings in Rotterdam, 56 in Hamburg, and 13 in Donostia-San Sebastian. Relative to k , that is about 1.4%, 1.6%, and 16.7% of prioritisation capacity respectively. Report borderline count, borderline share of N , and borderline count divided by k . This makes the Donostia result much more informative.

6. Strengthen the hazard perturbation experiment

Make the experimental sample and denominator explicit, especially because the Phase 2 borderline percentages differ strongly from the full Rotterdam comparison. More importantly, rerun each noise level

across multiple random seeds and report a distribution such as median with intervals rather than a single line from one perturbation realisation. This will show whether the observed non-monotonic pattern is structural or merely Monte Carlo noise.

7. Test sensitivity to the 0.2 / 0.8 stability thresholds

The current thresholds are defensible conventions, but still conventions. Add a compact sensitivity analysis using alternatives such as 0.1/0.9, 0.2/0.8, and 0.25/0.75. Keep the existing sensitivity to k . Also explain the useful identity that the sum of top- k membership probabilities across assets equals k , because every posterior draw contains exactly k selected assets. The meaningful object is therefore the width and uncertainty of the transition region.

8. Add a proper related-work section

The current reference list is too thin for a methodological preprint. Expand beyond Bayesian flood-risk models into Bayesian uncertainty propagation, ranking and selection uncertainty, decision-making under uncertainty, robust prioritisation, and possibly value-of-information or decision-focused modelling. Use this literature to position the contribution carefully and avoid unsupported novelty claims.

9. Reduce the visual bulk of the spatial results

The citywide and local maps are useful supporting evidence, but they currently consume a large fraction of the paper. Keep a compact three-city spatial comparison in the main text and move the large city-specific maps to an appendix or supplement. Use the recovered space for figures that make the statistical object tangible, for example posterior rank or membership behaviour for assets immediately around the decision boundary.

Scope discipline. Do not fold Phase 4, hydraulic validation, observed damage modelling, hierarchical latent hazard modelling, and a new operational system into this revision. The present paper is a methodological experiment about posterior decision stability. Its limitations should remain explicit rather than being hidden under additional machinery.

Recommended execution order

Rewrite abstract/introduction -> clarify synthetic DGP and experimental design -> fix Figure 1 -> rerun perturbations with replicates -> add borderline/ k and threshold sensitivity -> strengthen related work -> compress spatial figures -> final consistency and reproducibility pass.

Planning assumption: starting Thursday, a four-week revision window is plausible if the reruns and literature review stay contained. The main schedule risk is experimental expansion, not prose editing.